Matheus Hoffmann

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Summary

Senior ML engineer. I have worked on forecasting systems in production since 2017: churn prediction, fraud and loss forecasting, TV audience forecasting, and now ocean wave forecasting, where I make the WAVEWATCH III® model run on GPUs. I have worked in international teams since 2014, first on the broadcast operations of the FIFA World Cup 2014 and the Rio 2016 Olympics, and since 2020 remotely for companies in the UK and the US. I like to own a system end to end, from the raw data to the forecast that other teams use.

Experience

LabECO – Federal University of Santa Catarina — **Florianópolis, Brazil**
Wave Modeling Researcher — *Sep 2026 – present*

- Accelerating the WAVEWATCH III wave model (Fortran) on GPUs (NVIDIA H100) with C++/Kokkos and CUDA, with bit-for-bit parity against the Fortran, a Nix-pinned toolchain, Docker images and an agentic workflow built with Claude Code. Advisor: Professor Pedro Veras Guimarães. DOI 10.5281/zenodo.23221351.

TXP/ITV — **London, UK (remote)**
Senior ML/MLOps Engineer — *Jul 2025 – Aug 2026*

- Re-architected a single-channel audience forecast into a multi-channel, multi-model ML platform (ITV, ITV2, ITV3, ITV Quiz) with short, medium and long-term forecasting horizons.
- Owned the pipeline that runs 100+ models a day: data from internal APIs, via AWS Lambda, to Amazon S3; AWS Glue for ETL, Amazon SageMaker for training, MLflow for tracking, Airflow (MWAA) for pipeline orchestration; forecasts served to other teams from Amazon DynamoDB.

Signifyd — **San Jose, CA, USA (remote)**
Senior ML Engineer II (L5) — *Jun 2022 – Nov 2024*

- Cut loss-forecast experiment time by 93% (16 hours to 1 hour) by rewriting the simulation and moving it from pandas to PySpark. Later promoted to Subject Matter Expert (SME).
- Built anomaly detection on Databricks that identified organized fraud attacks, with the Airflow, dbt and Spark pipelines that feed it (Delta Lake, MLflow, PySpark MLlib).
- At Legiti, before its acquisition by Signifyd, cut the time and cost of batch feature computation by 77% by moving it from PostgreSQL to PySpark on Databricks, and helped build the fraud feature store (Delta Lake offline, Redis online).

Elemeno AI — **San Francisco, CA, USA (remote)**
Lead MLOps Engineer — *Jun 2021 – Jul 2022*

- Built an open-source MLOps platform on GKE for the retailer Americanas that took models from notebook to production: JupyterHub, Kubeflow Pipelines, Katib for tuning and KServe for containerized model serving, deployed with Terraform, Helm and ArgoCD.
- Customized JupyterHub, Elyra and MLflow so data scientists could run experiments on their own, and open-sourced the work with its documentation as three repositories at github.com/elemeno-ai.

Broad — **London, UK (remote)**
Senior Software Engineer — *Nov 2020 – Jun 2021*

- Helped take a Y Combinator fintech to its first closed beta, on the Flutter app and the Scala/ZIO backend: event sourcing on Kafka, plus the BIN sponsorship and FX integrations.

iU Pay / Guiabolso — **Rio de Janeiro and São Paulo, Brazil**
Software Engineer — *May 2019 – Nov 2020*

- Core contributor to Brazil's first instant cross-bank transaction engine, 8 months before Pix: Scala, Akka, Kafka event sourcing, Cassandra.

- **Stone** — **Rio de Janeiro, Brazil**
  Software Engineer — *Oct 2017 – Apr 2019*
  - Cut the logistics cost of changing the mobile carrier of point-of-sale (credit card) machines by 35%, with a recommendation model based on the signal quality of existing sellers and on proximity.
  - Migrated a churn prediction model from Bayesian networks to XGBoost on Spark, and built REST APIs and streaming services with Akka, deployed on Kubernetes.
- **LIOC – Ocean Instrumentation Lab, UFRJ** — **Rio de Janeiro, Brazil**
  Scientific Researcher Student — *Sep 2015 – Oct 2017*
  - FPGA signal processing in LabVIEW (NI cRIO-9068) for a Brazilian Navy underwater communication program, and real-time GPS tracking of boats on the TI MSP430, sending positions over 2G with mobile phone modules, with Python/PHP and MySQL.

Technical Writing and FOSS

- **marola** – open-source, nonprofit map of the best hour to swim at beaches in 3 Brazilian cities, from official bathing-water quality and sea forecasts. 13 public repositories, 77 written design proposals (MIPs), 6+ contributors in 3 countries. marola.dev, github.com/marola-dev.
- **nix-os** – NixOS configuration of 2 live VPSs that serve 5 public web services, among them the Hermes agent dashboard, an OpenRig instance and a metrics page. Every merge to main deploys both hosts with deploy-rs, and a host rolls back by itself if SSH stops answering. github.com/piabrota/nix-os.
- **gramatico** – Claude Code plugin, in Rust, that rewrites technical text in Brazilian Portuguese by level of rigor (v0.4.1, preprint on Zenodo). github.com/piabrota/gramatico.
- **pirsig** – second brain in Rust (Loco) that joins PARA and a Zettelkasten slip-box: a desktop app, a REST API and a CLI share one model layer. github.com/piabrota/pirsig.
- **Zenodo records (6 DOIs)**: marola 10.5281/zenodo.23224155; ww3-gpu 10.5281/zenodo.23221351; picasso 10.5281/zenodo.23311208; closed source 10.5281/zenodo.23333656; marola-bench 10.5281/zenodo.23352633; nashkell 10.5281/zenodo.23356739.

Skills

- **Machine learning**: forecasting, anomaly detection, recommendation systems, XGBoost, LightGBM, PySpark MLlib; LLM fine-tuning (QLoRA SFT, DPO), RAG, DSPy, local inference with Ollama and Ollaya
- **Serving and MLOps**: KServe, Kubeflow (Pipelines, Katib), Amazon SageMaker, MLflow, feature stores (Delta Lake offline, Redis online), Apache Airflow, Databricks, Spark, dbt
- **Infrastructure and reproducibility**: Docker, Kubernetes (EKS, GKE), Terraform, Helm, ArgoCD, Nix flakes, CI/CD (GitHub Actions, GitLab CI), AWS (S3, EC2, EMR, Glue, DynamoDB, Kinesis, Redshift)
- **Languages**: Python, Scala, C++, Fortran, TypeScript, SQL; Kafka, PostgreSQL, Cassandra
- **GPU, HPC and agentic coding**: CUDA, Kokkos, OpenMP, MPI; Claude Code (hooks, subagents, skills, MCP servers)

Education

- **Federal University of Rio de Janeiro (UFRJ)** — **Rio de Janeiro, Brazil**
  Engineering Degree, Electronics and Computer Engineering — *2013 – 2020*
- **Federal Center of Technological Education CEFET/RJ** — **Rio de Janeiro, Brazil**
  Electronics Technician — *2010 – 2012*

Certifications, Languages and Awards

- **AWS (4x)**: Machine Learning – Specialty; Machine Learning Engineer – Associate; Data Engineer – Associate; Cloud Practitioner. **PADI (4x)**: Open Water Diver; Advanced Open Water Diver; Enriched Air Nitrox; Rescue Diver.
- English (C1, full professional; 5+ years of remote work in English for UK and US companies), Portuguese (native), Spanish (fluent reading).
- **Signifyd "Values VIP: Tenacious"** – award for technical excellence, by employee vote (Feb 2024).
- **Oceanman Rio de Janeiro** – 3rd place, age group 30–39, 10 km open-water swim (Sep 2025).

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